**Statement of Work**
**Vehicle-to-Everything (V2X) Performance Measurement Support Services**

## I. BACKGROUND:

Intelligent Transportation Systems (ITS) incorporate vehicle-to-everything (V2X) communication technologies into roads and highways to improve the safety and efficiency of road transportation. ITS use a network of V2X devices to collect and analyze real-time traffic data to enable broadcasting highly relevant and localized information to road users for improved safety and efficiency. These V2X devices include both the onboard units (OBUs) in vehicles and the roadside units (RSUs) that comprise the digital road infrastructure. NIST's Communications Technology Laboratory (CTL) has a V2X Communications testbed that contains V2X devices from multiple vendors where research is conducted to accelerate the deployment of V2X technologies through better understanding of their technical capabilities and their feasible use cases given current vehicle features.

A significant barrier slowing deployment of V2X technologies is the interoperability of these devices. The idea of V2X communications spans a decade of innovation in communication technologies, leading to devices with diverse connectivity including Dedicated Short-Range Communications (DSRC), Long Term Evolution (LTE), and Fifth Generation (5G) New Radio (NR). From a deployment perspective, vehicle manufacturers will independently integrate OBUs into their products and each U.S. State will develop different RSU deployment strategies, creating a heterogenous landscape of devices that may never be tested against each other. There is a need for standard data formats and standardized test methodologies for ensuring the interoperability of V2X deployments.

## II. PURPOSE:

This Statement of Work focuses on the development of software tools and test methodologies for assessing device-to-device interoperability in support of the NIST V2X hardware testbed. This requires development of software to interface with commercial off-the-shelf hardware, configuration of test hardware, designing and executing interoperability tests, analyzing data captured from the testbed, and engagement with industry stakeholders through events such as plugfests.

This Statement of Work requires the following technical services:

(a) Conducting literature surveys and stakeholder outreach to inform NIST of the state of the art for interoperability testing of V2X devices.

(b) Designing and developing software to conduct interoperability tests using the hardware in the NIST V2X testbed.

(c) Collecting and analyzing data obtained from the NIST V2X testbed to assess the performance of V2X hardware.

(d) Supporting the hardware integration of new V2X devices and sensors into the NIST V2X testbed.

(e) Writing technical documents that describe the performed work.

Place of Performance:

NIST-Gaithersburg Campus.

Regular Business Hours:

Regular business hours are Monday through Friday, 8:00 am to 5:00 pm Local Time, excluding Federal holidays and NIST closures.

Period of Performance/Delivery Date:

06/01/2026 to 05/30/2027

III. SPECIFIC REQUIREMENTS:

Task 1 - Cellular V2X (C-V2X) Performance Measurement and Assessment

The contractor shall:

1.1 Conduct a survey of recent technical publications to identify relevant metrics and methodologies for assessing the performance of C-V2X devices that are agnostic to the device manufacturer.

1.2 Design and implement a software tool to assess the performance of V2X hardware in the NIST testbed based on the survey results. The software must be able to capture packets transmitted between different devices in the NIST testbed and analyze system performance using key performance indicators such as latency, packet loss, and bandwidth utilization.

1.3 Validate the software tool and assessment methodology using existing V2X data sets produced from the NIST V2X testbed.

1.4 Document the software tool and the performance assessment methodology it implements in a technical report for NIST research staff.

Task 2 - SCMS Interoperability Testing Use Cases

The contractor shall:

2.1 Coordinate with NIST research staff and industry stakeholders to catalogue the Security Credential Management System (SCMS) test methods in use by different SCMS providers and V2X device vendors.

2.2 Develop a framework for conducting SCMS interoperability testing including description of appropriate use cases and test scenarios to assess and verify IEEE 1609.2.1 interoperability.

2.3 Present the framework to industry stakeholders and make revisions based on their feedback to ensure alignment with the testing methodologies employed in industry-led plugfests.

Task 3 - NIST V2X Testbed Support

The contractor shall:

3.1 Integrate additional sensors such as LiDAR into the NIST V2X Testbed. The contractor shall integrate the sensor hardware into the testbed by mounting and wiring the sensors to the existing OBU/RSU devices, develop software for the OBU/RSU from multiple vendors to exchange data with the sensors, and develop test cases to demonstrate successful transmission of data.

3.2 Design and implement a use case where an RSU processes LiDAR data to transmit an appropriate message to approaching vehicles upon detection of an obstacle (such as a pedestrian) detected by the sensor. The messages exchanged must conform to existing SAE standards.

3.3 Investigate how the NIST V2X testbed could be integrated with the O-RAN testbed at NIST. The contractor shall coordinate with NIST research staff to understand and document the IT infrastructure between the testbeds, and shall produce an architectural diagram that describes a potential solution to transfer data between the two testbeds.

Task 4 - Travel

The Contractor may be required to travel to attend conferences and other meetings relevant to the tasks. NIST anticipates 1 domestic trip to an industry interoperability plugfest run by the OmniAir Consortium during the Period of Performance. This trip is expected to be less than a week in duration. NIST will provide at least three weeks' notice before each trip.

Prior approval from the Contracting Officer is required in these scenarios: lodging exceeding the government rate, airline/rental car upgrades, medical accommodations, driving instead of flying to destinations outside local travel, multiple destinations in a single trip, or trips longer than a week in duration.

Prior approval is requested by the contractor to the Contracting Officer via the TPOC. At a minimum, include the justification and documented cost comparison.

Task 5 - Safety-Related Equipment

The Contractor shall:

Provide all safety-related equipment required to perform the work under this award, including personal protective equipment (PPE) and other equipment that is required or appropriate to protect contractor personnel and other personnel on the NIST campus. Contractor-provided personal protective equipment shall include, at a minimum, hard hats, safety shoes, safety glasses (prescription or standard), respiratory protection (powered air purifying respirator or full-face mask respirator), high visibility clothing and gloves that meet applicable OSHA standards for PPE.

Task 6 - Training

The Contractor shall:

Provide safety-related training required by applicable OSHA, state and local regulations or standards to affected contractor personnel (including subcontractor personnel) that perform work under this award. Such training may include, for example, training for proper use of PPE, respirator use, fall protection, crane operation, forklift operation; electrical safety, or other safety-related topics.

Maintain records of safety training provided to contractor personnel (including subcontractor personnel) that perform work on this award. The contractor shall make those records available to the contracting officer, contracting officer's representative for the award, or other authorized personnel within 7 days of receipt of their written request, and annually thereafter or upon request.

Require contractor personnel (including subcontractors) to attend required NIST-specific health or safety training that is required by the award.

Task 7 - Health Hazard Assessments and Medical Evaluations

The Contractor shall:

Provide required health hazard exposure assessments, medical evaluations and screenings, and exposure monitoring required by applicable OSHA requirements.
Ensure PII information is protected.

Task 8 - Homeland Security Presidential Directive (HSPD)-12 (Required Task)

The Contractor (and/or any subcontractor) and its employees who require physical access to NIST facilities or logical access to NIST IT networks or systems must comply with Homeland Security Presidential Directive (HSPD)-12, Policy for a Common Identification Standard for Federal Employees and Contractors; OMB M-05-24; OMB M-19-17; FIPS 201, Personal Identity Verification (PIV) of Federal Employees and Contractors; NIST HSPD-12 policy; and Executive Order 13467, Part 1 §1.2.

The Contractor (and/or any subcontractor) must submit a roster by name, position, e-mail address, phone number and responsibility, of all staff working under this acquisition where the Contractor will develop, have the ability to access, or host and/or maintain a government information system(s).

The roster must be submitted to the COR and/or CO within 14 days of the effective date of this contract. Any revisions to the roster as a result of staffing changes must be submitted within 14 days of the change. The COR will notify the Contractor of the appropriate level of investigation required for each staff member. If the employee is filling a new position, the Contractor must provide a position description, and the Government will determine the appropriate suitability level.

Position Sensitivity Designations Requirement

All Contractor (and/or any subcontractor) employees must obtain a background investigation commensurate with their position sensitivity designation that complies with Parts 1400 and 731 of Title 5, Code of Federal Regulations (CFR).

The following position sensitivity designation levels apply to this solicitation/contact:

Low Risk

## IV. GENERAL REQUIREMENTS

Government Furnished Property

All property, data and information provided by the Government in the performance of this task remains the property of the Government and shall be surrendered to the government upon completion or termination of this requirement. Likewise, all deliverables generated under this requirement remain the property of the Government.

The property listed in this section (below) is property that is incidental in the place of performance due to the task order requiring the contractor personnel to be located on a Government site and the property used by the Contractor within the location remains accountable to the Government.  Per FAR Part 45.000(5) items considered to be incidental to the place of performance include office space, desk, chair, telephone computer and fax machine.

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|---|---|
| Item: | Laptop Computer |
| Model Number: | TBD |

Serial Number(s): TBD

Incidental Property Provided:

- Office Space
- Desk
- Chair
- Telephone
- Printer/Copy/Fax Machine

Upon award of this requirement, the Government will provide the Contractor with all access to all equipment and resources needed in the performance of this requirement.

REQUIREMENTS FOR CONTRACTOR PERSONNEL

(1) Qualifications: It is expected that this Statement of Work requires (1) full-time equivalent (FTE) Contractor Employee. Proposed key personnel shall have the following minimum education qualifications and experience:

(a) A master's degree from an accredited college or university in a field related to the work required

(b) At least 5 years of experience in distributed systems and the performance assessment of wireless networks

(c) Proficient in an object-oriented programming language such as Java

(d) Experienced with vehicle-to-everything communication technologies

(2) Safety: The Contractor employee must be responsible for knowing and complying with all installation safety prevention regulations. Such regulations include, but are not limited to, general safety, fire prevention, and waste disposal.

(3) Security: NIST is a restricted campus. An identification badge is required for access for entry into buildings and is shown to the armed Security Police when entering the campus.

(4) Identification Badges: Contractor employees must comply with NIST identification and access requirements. Each Contractor employee must wear a visible identification badge provided by the NIST Security Office.

(5) Vehicle Registration: All Contractor employees must register their vehicles with the NIST Security Office to gain access to the campus. A valid driver's license, Government-furnished civilian ID, proof of insurance and current registration must be presented to the NIST Security Office, at which time a NIST vehicle pass will be issued. The pass shall be displayed on the vehicle in accordance with NIST Security Office instructions.

V. Schedule Of Deliverables:

| Deliverable | Description | Format | Due Date | Quantity |
|---|---|---|---|---|
| 1 | Software that captures data from V2X devices from multiple vendors and uses key | Source Code (Python, etc.) | 9 months from date of award | 1 |

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| | performance indicators such as latency, packet loss, and bandwidth utilization to assess performance. | | | |
| 2 | Technical Report that describes the software tool produced from Task 1 including documentation of the test methodology, test metrics, and the software architecture. | MS Word or LaTeX Source | 12 months from date of award | 1 |
| 3 | Technical Report that describes a framework for conducting cross-vendor SCMS interoperability testing, including the assessment methodologies, testing requirements, and principal scenarios. | MS Word or LaTeX Source | 9 months from date of award | 1 |
| 4 | Software that integrates LiDAR data with V2X devices in the NIST testbed, including processing the data to trigger alerts upon detection of objects. | Source Code (Python, etc.) | 6 months from date of award | 1 |
| 5 | Technical Report that presents an architecture consistent with the NIST IT infrastructure for connecting the V2X testbed with other NIST test facilities. | MS Word or LaTeX Source | 12 months from date of award | 1 |
| 6 | Monthly Progress Report documenting: activities accomplished, deliverables completed, outstanding deliverables, reasons for delays. | MS Word | Monthly, before the 15th day of the next month | 12 |

Standards of Acceptance:

The NIST Point of Contact (POC) shall review all the above deliverables and respond with an acceptance or request for revision email to the Contractor POC within 14 days of receipt of deliverable. This section applies to deliverables which must be reviewed and approved for acceptances (e.g., draft plans, drawings, etc.)